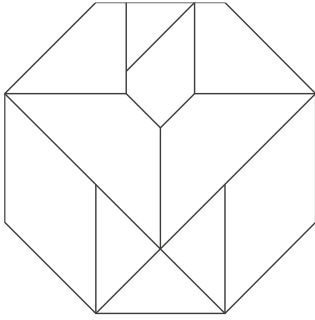




Smart Home Light Control App

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Alexandro Bolfa, student number 331500

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1. Introduction

Smart home lighting has become common in recent years, but many of the apps used to control these systems still feel harder to use than expected. In several cases, the navigation is not very intuitive, there are too many technical options on the screen, and the overall layout does not really fit the way people normally handle simple, everyday tasks. Because of this, the goal of this project was to design a smart home light control app that is both simple to use and enjoyable.

Throughout the project, the development process was guided by theories and principles taught during the User Experience and Usability course. One of the main ideas was that good design does not only need to be functional but should also support a positive emotional experience. Norman's emotional design framework: visceral, behavioral, and reflective levels, explains how users respond to products emotionally as well as practically. This approach helped guide many visual and interaction decisions in the prototype.

Another important foundation for the project was the Interaction Design process, which describes how designers move through an iterative cycle of understanding users, designing solutions, building prototypes, and evaluating them. This way of working allowed the prototype to evolve step by step based on real user needs and feedback.

The result is a high-fidelity Figma prototype that includes several key flows: adding homes, adding rooms, connecting devices, creating moods, and editing them. The aim was to create a system that feels logical, friendly, and supportive of the user's daily routines.

2. Methodology

The design process followed the four activities from the Interaction Design lifecycle: understanding users, creating design ideas, building prototypes, and evaluating them. This approach kept the project user-focused and helped guide each step of development.

The first stage of the project was aimed at figuring out who the app was meant for and what these users needed. Five people who already use smart lighting at home were interviewed to get a clearer picture (Appendix 1). During these talks, it became obvious that many existing lighting apps feel more complicated than expected. Several participants mentioned that the things they usually want to do are very straightforward, for example, turning a light on, changing a mood, or checking whether a device is connected. Even though the tasks themselves are simple, reaching them often requires clicking through several different menus and screens, which many found unnecessary and sometimes frustrating. The collected information enabled the creation of a new persona (Appendix 2) for the project, named Daniel, who represents the typical user profile identified in the interviews. He is 34 years old, comfortable with technology, and uses smart lighting every day to create different atmospheres at home. Daniel values efficiency and wants things to work without too much thought. This aligns with Norman's distinction between experiential and reflective cognition, which explains that frequently performed actions should feel intuitive and require minimal mental effort (Norman, 1993; Preece, Rogers & Sharp, 2019).

The next phase involved ideation and the development of initial concepts. A series of sketches was produced to explore possible ways of structuring the app. At this point in the process, the focus was mainly on keeping the app's structure easy to understand. The hierarchy, moving from Homes to Rooms to Devices and finally to Moods, needed to be clear so that users could find their way without much effort. To make the interface easier to use, the design relied more on recognition than on memory. In practice, this led to the use of simple icons, short labels, and small color previews, which made it easier for users to spot what they needed without having to remember anything. Studies on how people process information also suggest that grouped content and clear layouts are understood much faster than long blocks of text, which makes this approach well aligned with how users usually take in visual information (Tullis, 1987; Weller, 2004; Preece, Rogers & Sharp, 2019). Alongside these choices, the app's visual style was developed with inspiration from Norman's emotional design ideas. Soft gradients, rounded

edges, and warm color tones were selected to create a calm and approachable appearance, helping to establish a positive first impression and smoother user interaction.

Once the main concept was in place, the project moved into the prototyping stage. The design work started out with simple, low-fidelity sketches and then became more detailed over time, as suggested by Preece, Rogers, and Sharp (2019). Building it up gradually made it possible to test ideas early on and make changes along the way as the direction of the design became clearer. This prototype has been implemented in Figma and covered all essential flows, including adding homes, adding rooms, connecting devices, creating moods, and editing existing ones. The system for adjusting light colors underwent several iterations. Early versions offered only basic controls, but these were expanded into a full Hue, Saturation, Value configuration, allowing users to access a full range of color options.

The final stage consisted of evaluating the prototype. Usability testing (Appendix 3) was conducted using the Think-Aloud method, in which participants performed tasks while verbalizing their thoughts. To reduce the risk of influencing participant behavior, the test followed the improved moderation approach described in the CHI 2010 study, which recommends brief and neutral prompts when participants fall silent. This made it possible to notice moments where users hesitated or seemed unsure about what to do next. The usability test was put together using the task-focused evaluation and observation principles described in the Interaction Design textbook. To support this, the session was also moderated using the improved think-aloud approach outlined by Olmsted-Hawala et al. (2010), which helped capture users' thoughts without interrupting the flow of the test.

In total, five participants were involved in the evaluation, and their feedback had a significant impact on the adjustments that shaped the final version of the prototype.

3. Findings

The findings were developed gradually through the iterative work carried out within the four main activities of the Interaction Design process.

The interviews carried out early in the project showed that users generally preferred a lighting app that felt simple to use rather than overly technical or heavy with settings. Many participants mentioned that they wanted to move through the app quickly, without having to open long or complicated menus. Tasks

such as adding homes or rooms were expected to be straightforward, and the process of connecting devices needed to be clear enough that users would not have to stop and think too much. When adjusting moods, visual feedback played an important role because it allowed users to see the effect of their changes right away. Several people also commented that they liked interfaces that felt calm and welcoming, as opposed to those that appeared busy or overwhelming.

During the design phase, these preferences were translated into specific interface choices. Icons and short labels were included to make each section easier to understand briefly. Color previews were added so that mood adjustments would be visible immediately. The layout also relied on spacing and grouping to organize the information, which helped users grasp the structure of each screen more quickly.

The prototyping phase allowed these ideas to become real, interactive screens. Each flow was tested and adjusted several times. The Add Home (Appendix A) and Add Room (Appendix B) flows follow the same structure, making them easier to learn. The Connect Device flow (Appendix C) uses a step-by-step approach so that the user sees only what they need at each moment. The Add Mood (Appendix D) flow starts with all devices turned off, encouraging users to build a mood from scratch. In the Edit Mood flow (Appendix E), changes are saved automatically, and users can see the AI-generated image update in real time.

During the evaluation stage, all participants were able to complete the tasks, although a few parts still needed small adjustments. Most people got through the steps for adding homes and rooms without any real trouble, and nothing in those parts seemed to confuse them. The device-connection step also went well, although a few participants slowed down a bit while working through it. The only real pause happened when they had to choose a room for a device that had just been connected. A few users paused at that step for a moment, trying to understand what the app expected them to do before they continued. The mood-screen creation screens received the most positive reactions, and several users mentioned that the color sliders were easy to understand and quite fun to play around with.

However, two major issues have appeared. First, the save button for creating moods was too small (Appendix F) and looked too much like a decorative icon, so some users did not notice it immediately. After enlarging it and adding a clearer label (Appendix G), the issue disappeared in the next test. Second, the Rooms/Devices switch (Appendix H) at the top of the screen was not immediately obvious to all

users. Some expected it to be more visually distinct. Overall, the tests confirmed that the interface was understandable, visually appealing, and efficient once the minor usability issues were resolved.

4. Discussion

Thinking back on the design work, it was obvious that emotional design ideas, basic cognitive principles, and user-centered approaches all played a role in shaping the final app. Norman's three levels of emotional design were also relevant throughout this process. The visceral side of the design, things like the soft gradients, calmer color tones, and the uncomplicated icons, helped create an interface that users generally felt comfortable using. A couple of participants even referred to the design as "relaxing," which lines up with Norman's view that positive emotions can make people more willing to explore and more patient when interacting with technology.

At the behavioral level, the use of consistent layouts, simple navigation, and quick feedback made it easier for users to move through the app without having to think too much about each step. The interaction flows require very little mental effort, which aligns with the distinction between experiential and reflective cognition described by Norman. Experiential cognition refers to intuitive, fast, and effortless action patterns, making it well suited for tasks that users perform repeatedly (Norman, 1993; Preece, Rogers & Sharp, 2019). Since people tend to adjust their lights several times a day, the design needed to support fast and effortless interactions.

At the reflective level, the app offered several ways for users to personalize their setup, for example by naming their moods, picking their own colors, or looking at the AI-generated previews. These features made the system feel more personal and gave users a stronger sense of involvement in how their lighting was arranged. This matches Norman's view that products become more meaningful when they express something about the user's identity or creativity.

The usability tests highlighted several strong points in the prototype but also brought attention to areas that needed improvement. The Think-Aloud method made it easier to notice moments where participants hesitated or seemed unsure, although it also revealed the challenge noted in the CHI 2010 paper: participants often fall silent unless they are gently encouraged to keep talking. Even so, the feedback

gathered during testing was helpful. The problems with the save button and the Rooms/Devices switch were identified early and adjusted in later iterations.

There were still limitations in the design. One example is the “device linking” function that appeared in one of the last versions of the prototype. The idea was to let users link several lights so they would follow the same color settings (Appendix I). Although the feature was useful in theory, the additional controls made the interface feel more complicated during testing. Because of this, the feature was removed to keep the experience clearer. It may still be worth considering again later, but in a simpler form.

Overall, the project highlighted the value of balancing usability with the emotional side of design. A system that focuses only on functionality may perform its tasks, but it does not automatically create a pleasant or trustworthy experience. Bringing together visual design choices, straightforward interaction patterns, and what was learned from user testing helped shape a prototype that feels both practical and engaging to use.

5. Conclusion

This project demonstrated how user-centered design, emotional design, and cognitive principles can work together to create a smart-home lighting app that is easy to use and enjoyable. Through several iterations of user understanding, solution design, prototyping, and testing, the final interface now supports clear navigation, personalized mood creation, and efficient device management across multiple homes and rooms.

The findings indicate that emotional design contributes not only to the visual appeal of the app but also to how smoothly it can be used. Applying cognitive principles helped lower the mental effort required from users, which made frequent tasks quicker and easier to carry out. The Think-Aloud usability tests also played an important role, as they brought several issues to light early in the process, allowing the interaction flow to be adjusted and improved before moving on.

If the project were to continue, one possible direction would be to take another look at linking several devices so they can work together. It could also be useful to improve accessibility options or try the app with a larger and more diverse group of users. Eventually, the work could move toward building an actual

working version of the design. Overall, the project showed how useful an iterative design process can be and how the theory from the course can support practical design decisions.

6. Appendix

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Appendix 1 – Interview summaries.pdf

Appendix 2 – Persona.pdf

Appendix 3 – Usability tests.pdf

Appendix A - Prototype 5 Add Home Flow.png

Appendix B - Prototype 5 Add Room Flow.png

Appendix C - Prototype 5 Connect Device Flow.png

Appendix D - Prototype 5 Add Mood Flow.png

Appendix E - Prototype 5 Edit Mood Flow.png

Appendix F - Prototype 1 Mood Screen & Side Menu Open Screen & Add New Mood Screen.png

Appendix G - Prototype 2 Mood Screen & Side Menu Open Screen & Add New Mood Screen.png

Appendix H - Prototype 3 Home List Screen - Room List Screen - Device List Screen.png

Appendix I - Light Configuration Versions.png